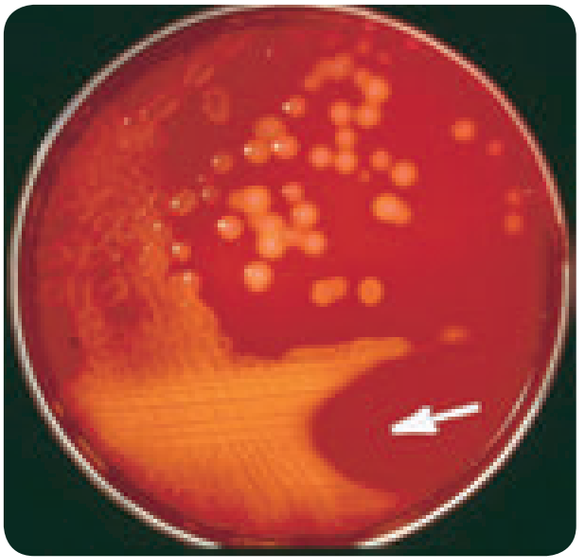


Figure 7.5: Colony morphology of *Streptococcus pyogenes* on blood agar

7.4.3 Antigenic Structure

Capsule: It inhibits phagocytosis

Cell wall: The outer layer of cell wall consists of protein and lipoteichoic acid which helps in attachment to the host cell. Middle layer of cell wall consists of Group Specific C – Carbohydrate that is used for Lancefield grouping. Inner layer of cell wall is made up of peptidoglycan which has pyrogenic and thrombolytic activity.

Toxins and Enzymes: *Streptococcus pyogenes* produces several exotoxins and enzymes which contribute to its virulence.

Toxins and Hemolysins: *Streptococci* produces two types of hemolysins which are Streptolysin O and Streptolysin S.

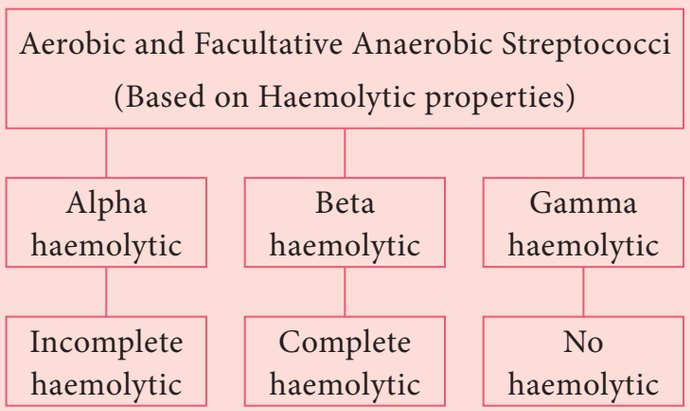
Erythrogenic toxin: (Pyrogenic exotoxin) – The induction of fever is the primary effect of this toxin and it is responsible for the rash of scarlet fever.

Enzymes: The various enzyme of *Streptococcus pyogenes*which exhibits virulence activity are listed in Table 7.3.

Infobits

Classification

The aerobic and facultative anaerobic *Streptococcus* are classified based on haemolytic properties. Three types of haemolytic reactions are observed on blood agar medium, which are:



Based on serological grouping of carbohydrate C antigen of Beta haemolytic organisms, they are classified into 20 Lancefield groups (A to H & K to V)

Streptococcus pyogenes is beta haemolytic organism which is included in Group A.

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